

Lecture 4: Expectation

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Overview

- 1 Expectation & Variance
- 2 Geometric and Negative Binomial
- 3 Indicator R.V.s and The Fundamental Bridge
- 4 Moments and Indicators
- 5 Poisson
- 6 Distance between Two Probability Distributions
- 7 Probability Generating Functions
- 8 Reading for Fun

Outline

- 1 Expectation & Variance
- 2 Geometric and Negative Binomial
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Expectation of A Discrete R.V.

Definition

The *expected value* (also called the *expectation* or *mean*) of a discrete r.v. X whose distinct possible values are $x_1, x_2, \dots$ is defined by

$$E(X) = \sum_{j=1}^{\infty} x_j P(X = x_j).$$

If the support is finite, then this is replaced by a finite sum. We can also write

$$E(X) = \sum_x \underbrace{x}_{\text{value}} \underbrace{P(X = x)}_{\text{PMF at } x},$$

where the sum is over the support of X .

Distribution

if $E(X) = E(Y) \not\Rightarrow X \sim Y$.

$$X = \begin{cases} 100, & \text{w.p. } 1/2 \\ 0, & \text{w.p. } 1/2 \end{cases}$$

$$Y = \begin{cases} 20, & \text{w.p. } 1/2 \\ 30, & \text{w.p. } 1/2 \end{cases}$$

Theorem

If X and Y are discrete r.v.s with the same distribution, then $E(X) = E(Y)$ (if either side exists).

$$E(X) = 50 \quad \underline{\underline{=}} \quad E(Y) = 50.$$

Linearity

The expected value of a sum of r.v.s is the sum of the individual expected values.

Theorem

For any r.v.s X, Y and any constant c ,

$$E(X + Y) = E(X) + E(Y),$$

$$E(cX) = cE(X).$$

Monotonicity of Expectation

$$Z = X - Y \Rightarrow Z \geq 0, \text{ w.p. } 1.$$

$$\Rightarrow E(Z) \geq 0 \Rightarrow E(X - Y) \geq 0$$

$$\Rightarrow E(X) - E(Y) \geq 0$$

Theorem

Let X and Y be r.v.s such that $X \geq Y$ with probability 1. Then $E(X) \geq E(Y)$, with equality holding if and only if $X \equiv Y$ with probability 1.

$$\Rightarrow E(X) \geq E(Y).$$

Expectation via Survival Function



Theorem

Let X be a nonnegative integer-valued r.v. Let F be the CDF of X , and $G(x) = 1 - F(x) = P(X > x)$. The function G is called the survival function of X . Then

Tail distribution

$$E(X) = \sum_{n=0}^{\infty} G(n).$$

$$= \sum_{n=0}^{\infty} P(X > n).$$

$$= \sum_{n=1}^{\infty} P(X \geq n).$$

That is, we can obtain the expectation of X by summing up the survival function (or, stated otherwise, summing up tail probabilities of the distribution).

Proof

$$m \geq n \geq 1.$$

$$\begin{aligned} \sum_{n=0}^{\infty} G(n) &= \sum_{n=1}^{\infty} \underbrace{P(X \geq n)} \\ &= \sum_{n=1}^{\infty} \left[\sum_{m=n}^{\infty} P(X=m) \right] \end{aligned}$$

Fubini Theorem:

$$= \sum_{m=1}^{\infty} \sum_{n=1}^m P(X=m).$$

$$= \sum_{m=1}^{\infty} m \cdot P(X=m)$$

$$+ 0 \cdot P(X=0)$$

$$= E(X).$$

$$P(X \geq 1) = P(X=1) + P(X=2) + P(X=3) + \dots$$

$$+ P(X \geq 2) = P(X=2) + P(X=3) + \dots$$

$$+ P(X \geq 3) = P(X=3) + \dots$$

$$+ \vdots$$

$$\vdots$$

$$1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + \dots$$

$$= \sum_{n=1}^{\infty} n \cdot P(X=n) + 0 \cdot P(X=0)$$

$$= \sum_{n=0}^{\infty} n \cdot P(X=n) = E(X).$$

Law Of The Unconscious Statistician (LOTUS)

Theorem

If X is a discrete r.v. and g is a function from $\mathbb{R}$ to $\mathbb{R}$, then

$$E(g(X)) = \sum_x g(x)P(X = x),$$

where the sum is taken over all possible values of X .

$$E(e^{tx}) = \sum_x e^{tx} P(X=x).$$

Variance and Standard Deviation

	$\frac{1}{2}$	$\frac{1}{2}$
X :	49	51
Y :	0	100

$$E(X) = 50 = E(Y).$$

$$\text{Var}(X) = 1 \quad \text{Var}(Y) = 2500$$

Definition

The variance of an r.v. X is

$(x - EX)^2$ distance.

$$\text{Var}(X) = E((X - EX)^2).$$

The square root of the variance is called the *standard deviation (SD)*:

$$SD(X) = \sqrt{\text{Var}(X)}.$$

Properties of Variance

- ① Let $\mu = E(X)$. $\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E(X^2) - \mu^2$.
- ② $Y = X + c$ $E(Y) = E(X) + c$. $E[(Y - E(X) - c)^2] = E[(X - E(X))^2]$
- ③ $Y = cX$, $E(Y) = cE(X)$.
- ① • For any r.v. X , $\text{Var}(X) = E(X^2) - (EX)^2$.
- ② • $\text{Var}(X + c) = \text{Var}(X)$ for any constant c .
- ③ • $\text{Var}(cX) = c^2 \text{Var}(X)$ for any constant c .
- ④ • If X and Y are independent, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. $E(XY) = E(X)E(Y)$
- ⑤ • $\text{Var}(X) \geq 0$ with equality if and only if $P(X = a) = 1$ for some constant a .
- ④ $\text{Var}(X + Y) = E[(X + Y - (EX + EY))^2] = E\{[X - E(X) + Y - E(Y)]^2\}$
 $= E\{[X - E(X)]^2 + 2[X - E(X)][Y - E(Y)] + [Y - E(Y)]^2\}$
 $= \text{Var}(X) + 2E[X - E(X)]E[Y - E(Y)] + \text{Var}(Y)$

Properties of Variance

$$\textcircled{5} \quad \text{Var}(X) \geq 0 \quad \Leftrightarrow \quad E[(X - EX)^2] \geq 0.$$

$$\Leftrightarrow E(X^2) - (EX)^2 \geq 0$$

$$E(X^2) \geq (EX)^2.$$

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Story: Geometric Distribution

$$k \geq 0.$$

$$\underline{P(X=k) = (1-p)^k \cdot p.}$$

$k+1$ Bernoulli trials.

first k trials failure,
the $(k+1)$ -th trial success.

Consider a sequence of independent Bernoulli trials, each with the same success probability $p \in (0, 1)$, with trials performed until a success occurs. Let X be the number of **failures** before the first successful trial. Then X has the Geometric distribution with parameter p ; we denote this by $X \sim \text{Geom}(p)$.

Geometric PMF

$$\textcircled{1} E(X) = \sum_{k=0}^{\infty} k \cdot P(X=k) = \sum_{k=0}^{\infty} k \cdot q^k \cdot p = p \cdot \sum_{k=0}^{\infty} k \cdot q^k = \frac{q}{p} = \frac{1-p}{p}$$

$$\textcircled{2} P(X \geq 0) = 1. \quad k \geq 1.$$

$$P(X \geq k) = 1 - P(X < k) = 1 - P(X \leq k-1) = 1 - \sum_{j=0}^{k-1} P(X=j) = 1 - \sum_{j=0}^{k-1} q^j p$$

Theorem

If $X \sim \text{Geom}(p)$, then the PMF of X is

$$P(X = k) = q^k p$$

for $k = 0, 1, 2, \dots$, where $q = 1 - p$.

$$= 1 - p \frac{1 - q^k}{1 - q} = q^k$$

$$E(X) = \sum_{k=0}^{\infty} P(X > k) = \sum_{k=1}^{\infty} P(X \geq k) = \frac{1-p}{p} = \sum_{k=1}^{\infty} q^k$$

Memoryless Property

$$1^\circ k=0, \quad P(X \geq n | X \geq 0) = P(X \geq n).$$

$$2^\circ k \geq 1, \quad P(X \geq n+k | X \geq k) = \frac{P(X \geq n+k, X \geq k)}{P(X \geq k)} = \frac{P(X \geq n+k)}{P(X \geq k)} = \frac{\varrho^{n+k}}{\varrho^k}$$

Theorem

If $X \sim \text{Geom}(p)$, then for any positive integer n ,

$$= \varrho^n.$$

$$P(X \geq n+k | X \geq k) = P(X \geq n)$$

for $k = 0, 1, 2, \dots$

Memoryless Property

$$1^\circ P(X \geq n+k | X \geq k) = \frac{P(X \geq n+k)}{P(X \geq k)} = P(X \geq n) \Rightarrow \underline{P(X \geq n+k) = P(X \geq n)P(X \geq k)}$$

$$2^\circ k=0; P(X \geq n) = P(X \geq n)P(X \geq 0) \Rightarrow P(X \geq 0) = 1.$$

$$3^\circ \underline{G(n) = P(X \geq n)}; G(0) = 1.$$

Theorem

Suppose for any positive integer n , discrete random variable X satisfies

$$P(X \geq n+k | X \geq k) = P(X \geq n) \quad \begin{matrix} G(1) = P(X \geq 1) \leq 1. \\ \text{Let } 0 \leq q = G(1) \leq 1. \end{matrix}$$

non-negative integer,
for $k = 0, 1, 2, \dots$, then $X \sim \text{Geom}(p)$.

$$4^\circ \underline{G(n+k) = G(n) \cdot G(k)}, \quad n=k=1. \quad G(2) = G^2(1) = q^2.$$

$$G(n) = q^n \Rightarrow P(X \geq n) = q^n.$$

$$\Rightarrow P(X=0) = 1 - P(X \geq 1) = 1 - q.$$

$$P(X=k) = P(X \geq k-1) - P(X \geq k) = q^{k-1} - q^k = q^{k-1}(1-q)$$

Memoryless Property

Theorem

Geometric distribution is the one and the only one discrete distribution that is memoryless.

First Success Distribution

$$X \sim \text{Geom}(p)$$

$$P(X=k) = q^k \cdot p$$

$$Y \sim \text{FS}(p)$$

$$Y = X + 1$$

$$P(Y=k) = q^{k-1} \cdot p$$

Definition

In a sequence of independent Bernoulli trials with success probability p , let Y be the number of trials until the first successful trial, including the success. Then Y has the First Success distribution with parameter p ; we denote this by $Y \sim \text{FS}(p)$.

Example: Geometric & First Success Expectation

$$1^{\circ} E(X) = \frac{1-p}{p}$$

$$2^{\circ} E(Y) = E(X+1) = 1 + E(X) = \frac{1}{p}.$$

Let $X \sim \text{Geom}(p)$ and $Y \sim \text{FS}(p)$, find $E(X)$ and $E(Y)$.

Story: Negative Binomial Distribution

$$P(X=n): \text{ } n \text{ failures before the } r\text{-th success}$$

$$(n+r) \text{ trials.}$$

$$\binom{n+r-1}{n} (1-p)^n \cdot p^r$$

$$\text{the } (n+r)\text{-th trial is successful.}$$

In a sequence of independent Bernoulli trials with success probability p , if X is the number of failures before the r^{th} success, then X is said to have the Negative Binomial distribution with parameters r and p , denoted $X \sim \text{NBin}(r, p)$.

$(n+r-1)$ trials, $(r-1)$ success
 n failures.

$$\binom{n+r-1}{n} (1-p)^n \cdot p^{r-1}$$

Negative Binomial PMF

$$\textcircled{1} \binom{-r}{n} = \frac{(-r)!}{n!(-r-n)!} = \frac{(-r-n+1)(-r-n+2)\cdots(-r)}{n!} = \frac{(-r)(-r-1)\cdots(-r-n+1)}{n!}$$

$$\textcircled{2} \binom{n+r-1}{r-1} = \frac{(n+r-1)!}{n!(r-1)!} = \frac{\overbrace{r(r+1)\cdots(n+r-1)}^{n!}}{n!} = (-1)^n \binom{-r}{n}$$

Theorem

If $X \sim \text{NBin}(r, p)$, then the PMF of X is

$$P(X = n) = \binom{n+r-1}{r-1} p^r q^n \quad \text{Valid PMF}$$

for $n = 0, 1, 2, \dots$, where $q = 1 - p$.

$$\begin{aligned} \sum_{n=0}^{\infty} P(X=n) &= 1 & p^{-r} &= (1-q)^{-r} = \sum_{n=0}^{\infty} \binom{-r}{n} (-q)^n 1^{-r-n} = \sum_{n=0}^{\infty} \binom{-r}{n} (-q)^n \\ & & &= \sum_{n=0}^{\infty} \binom{n+r-1}{r-1} q^n \\ 1 &= \sum_{n=0}^{\infty} \binom{n+r-1}{r-1} q^n p^r = \sum_{n=0}^{\infty} P(X=n). \end{aligned}$$

Geometric & Negative Binomial

$X_1 \sim \text{Geom}(p)$ X_1 : # of failures before the 1st success. *memoryless*

$X_2 \sim \text{Geom}(p)$ X_2 : # between the 1st and 2nd success.

$X_3 \sim \text{Geom}(p)$ X_3 : # 2nd and 3rd . . .

Theorem

Let $X \sim \text{NBin}(r, p)$, viewed as the number of failures before the r^{th} success in a sequence of independent Bernoulli trials with success probability p . Then we can write $X = X_1 + \dots + X_r$ where the X_i are i.i.d. $\text{Geom}(p)$.

$X_i \sim \text{Geom}(p)$ X_i : $(i-1)^{\text{th}}$. . . i^{th} . . .

Example: Expectation

Method 1: $E(X) = \sum_{n=0}^{\infty} n \cdot P(X=n) = \sum_{n=0}^{\infty} n \cdot \binom{n+r-1}{r-1} \cdot p^r \cdot q^n.$

Method 2: $X = X_1 + X_2 + \dots + X_r. \quad X_i \sim \text{Geom}(p).$

Let $X \sim \text{NBin}(r, p)$, find $E(X)$.

$$E(X) = \underline{E(X_1)} + E(X_2) + \dots + E(X_r) = r \cdot \frac{1-p}{p}.$$

Example: 小浣熊干脆面与水浒英雄卡



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Example: 小浣熊干脆面与水浒英雄卡

为了收集齐108张水浒英雄卡，平均而言你需要购买多少包小浣熊方便面？

Example: 盲盒收集



Example: 盲盒收集



Model: Coupon Collector

Suppose there are n types of toys, which you are collecting one by one, with the goal of getting a complete set. When collecting toys, the toy types are random (as is sometimes the case, for example, with toys included in cereal boxes or included with kids' meals from a fast food restaurant). Assume that each time you collect a toy, it is equally likely to be any of the n types. Let N denote the number of toys needed until you have a complete set. Find $E(N)$ and $Var(N)$.

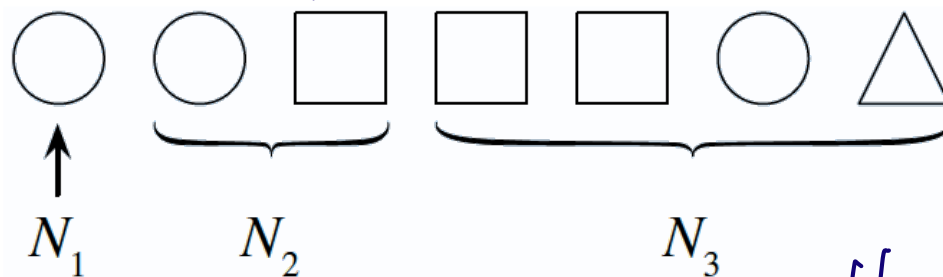
Solution: Coupon Collector

1° N : # of toys needed to obtain all types of toys.

$$N = N_1 + N_2 + \dots + N_n. \quad \text{Memoryless}$$

N_1 : # of toys until the first new toy type appeared. $N_1 = 1$.

N_2 : additional # of toys until the second new toy type appeared.



N_n : n -th

$$2^\circ N_2 \sim \text{FS}\left(1 - \frac{1}{n}\right), \quad N_3 \sim \text{FS}\left(1 - \frac{2}{n}\right), \quad N_j \sim \text{FS}\left(1 - \frac{j-1}{n}\right)$$

$$3^\circ E(N) = E(N_1) + E(N_2) + \dots + E(N_n) \quad \text{FS}\left(\frac{n-j+1}{n}\right)$$

Solution: Coupon Collector

$$E(N) = E(N_1) + E(N_2) + \dots + E(N_n)$$

$$= 1 + \frac{n}{n-1} + \dots + n.$$

$$= n \left(\frac{1}{n} + \frac{1}{n-1} + \dots + 1 \right) = n \cdot \sum_{j=1}^n \frac{1}{j} \quad n \gg 1$$

$$\approx n \cdot (\ln n + 0.577).$$

$$n = 10^8. \quad \approx 567.98.$$

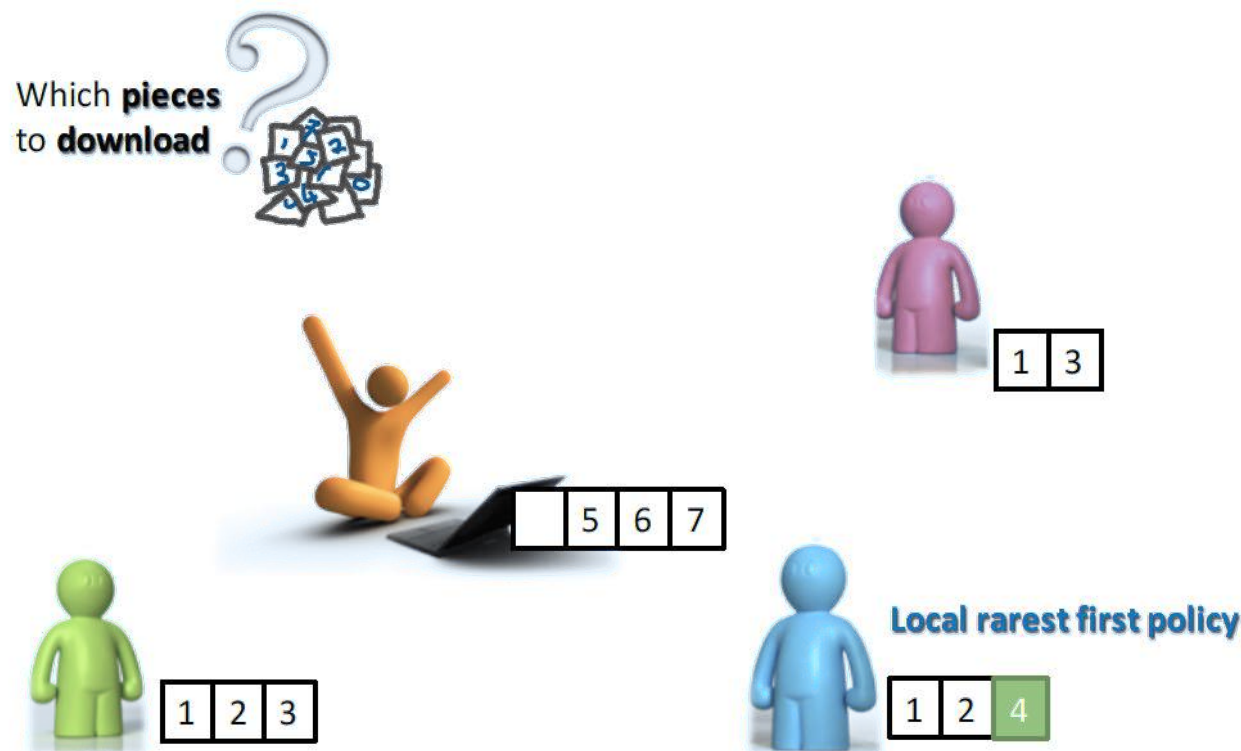
$$4^\circ \text{ Var}(N) = \text{Var}(N_1) + \text{Var}(N_2) + \dots + \text{Var}(N_n).$$

Application: Peer-to-Peer System

- Target file is decomposed into n pieces.
- Each peer randomly downloads pieces and uploads pieces from its neighbors.
- $\Theta(n \ln n)$ downloads to complete the downloading file.
- The last block problem: missing the last piece (stop at 99% downloading progress).

Application: Peer-to-Peer System

- Solution adopted by BitTorrent:
 - ▶ tries to download a block that is least replicated among its neighbors.
 - ▶ maximize the diversity of content in the system, i.e., make the number of replicas of each block as equal as possible.



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Properties of Indicator R.V.

$$A^c = \bar{A}$$

$$I_A = \begin{cases} 1, & \text{if event } A \text{ occurs.} \\ 0, & \text{otherwise.} \end{cases}$$

Let A and B be events. Then the following properties hold.

① $(I_A)^k = I_A$ for any positive integer k . $1^k = 1, 0^k = 0$.

② $I_{A^c} = 1 - I_A$. ③ $I_{A \cap B} = 1 \Rightarrow A \cap B$ occurs. Both A and B occurs.

③ $I_{A \cap B} = I_A I_B$. $I_A = 1, I_B = 1$

④ $I_{A \cup B} = I_A + I_B - I_A I_B$.

$I_{A \cap B} = 0$, $A \cap B$ not occurs

$I_A = 0$, or $I_B = 0$.

$$I_{A \cup B} \stackrel{②}{=} 1 - I_{\overline{A \cup B}}$$

$$= 1 - I_{A^c \cap B^c}$$

$$\stackrel{②}{=} 1 - I_{A^c} I_{B^c}$$

$$\stackrel{②}{=} 1 - (1 - I_A)(1 - I_B)$$

$$= I_A + I_B - I_A I_B$$

Fundamental Bridge Between Probability and Expectation

$$I_A = \begin{cases} 1, & \text{if event } A \text{ occurs, } P(A). \\ 0, & \text{otherwise, } 1 - P(A). \end{cases}$$

$$E(I_A) = 1 \cdot P(A) + 0 \cdot (1 - P(A)) = P(A).$$

Theorem

There is a one-to-one correspondence between events and indicator r.v.s, and the probability of an event A is the expected value of its indicator r.v. I_A :

$$P(\underline{A}) = \underline{E(I_A)}.$$

union
set operation

low complexity

Example: Boole's Inequality

$$\langle 1 \rangle \quad I(A_1 \vee A_2 \vee \cdots \vee A_n) \leq I(A_1) + I(A_2) + \cdots + I(A_n). \quad \checkmark$$

$$1^\circ \text{ if } LHS = 0; \quad RHS \geq 0. \quad \checkmark$$

$$2^\circ \text{ if } LHS = 1, \Rightarrow \text{at least one } A_j \text{ occurs. } \exists j, I(A_j) = 1.$$

For any n events $A_1, A_2, \dots, A_n$,

$$\Rightarrow RHS \geq 1.$$

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

$\langle 2 \rangle$ Take expectation for both sides.

Solution: Booler's Inequality

Example: Inclusion-Exclusion Formula

$$1^\circ I(A_1 \cup A_2 \cup \dots \cup A_n) = 1 - \overline{I(A_1 \cup A_2 \cup \dots \cup A_n)}$$

$$\overline{I(A_1 \cup A_2 \cup \dots \cup A_n)} = I(\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}) = I(A_1^c) I(A_2^c) \dots I(A_n^c)$$

$$\text{For any events } A_1, \dots, A_n, \quad = [1 - I(A_1)] [1 - I(A_2)] \dots [1 - I(A_n)]$$

$$= 1 - \sum_i I(A_i) + \sum_{i < j} I(A_i) I(A_j) + \dots + (-1)^{n+1} I(A_1) \dots I(A_n)$$

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n).$$

$$2^\circ I(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_i I(A_i) - \sum_{i < j} I(A_i) I(A_j) + \dots + (-1)^{n+1} I(A_1) \dots I(A_n)$$

3° Take expectations for both sides.

Solution: Inclusion-Exclusion Formula

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Moments of Indicator Methods

- Given n events $A_1, \dots, A_n$ and indicators $I_j, j = 1, \dots, n$.
 - $X = \sum_{j=1}^n I_j$: the number of events that occur.
 - $\binom{X}{2} = \sum_{i < j} I_i I_j$: the number of pairs of distinct events that occur.
 $I_i I_j = I(A_i)I(A_j) = I(A_i \cap A_j)$
 - $E\left(\binom{X}{2}\right) = \sum_{i < j} P(A_i \cap A_j) \Rightarrow E\left[\frac{X(X-1)}{2}\right] = \sum_{i < j} P(A_i \cap A_j)$
 - $E(X^2) = 2 \sum_{i < j} P(A_i \cap A_j) + E(X)$.
 - $\text{Var}(X) = 2 \sum_{i < j} P(A_i \cap A_j) + E(X) - (E(X))^2$.
- $$\text{Var}(X) = E(X^2) - (EX)^2$$

Moments of Binomial Random Variables

$$X \sim \text{Bin}(n, p);$$

1° consider n independent Bernoulli trials, each with success prop. p .

event A_i : the i -th trial is a success. $P(A_i) = p, \forall 1 \leq i \leq n$.

$$I_j = I(A_j) \sim \text{Bern}(p), \quad \forall 1 \leq j \leq n.$$

2° # of successful trials. $X = \sum_{j=1}^n I_j$.

$$\textcircled{1} E(X) = E\left(\sum_{j=1}^n I_j\right) = np.$$

$$\textcircled{2} E\left[\binom{X}{2}\right] = \sum_{i < j} P(A_i \cap A_j) = \sum_{i < j} P(A_i) P(A_j) = \binom{n}{2} \cdot p^2 = \frac{n(n-1)}{2} p^2$$

$$E(X^2) = 2 \sum_{i < j} P(A_i \cap A_j) + E(X) = n(n-1)p^2 + np.$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = np(1-p)$$

$$\textcircled{3} E\left[\binom{X}{k}\right] = \binom{n}{k} p^k.$$

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Poisson Distribution

① Valid PMF $\sum_{k=0}^{\infty} P(X=k) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \stackrel{e^{\lambda} \text{ Taylor expansion}}{=} 1.$

Definition

An r.v. X has the *Poisson distribution* with parameter λ if the PMF of X is

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, k = 0, 1, 2, \dots$$

We write this as $X \sim \text{Pois}(\lambda)$.

$$E(X) = \text{Var}(X) = \lambda.$$

Example: Poisson Expectation & Variance

$$\textcircled{1} E(X) = \sum_{k=0}^{\infty} k \cdot P(X=k) = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \cdot \lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda} \cdot \lambda^{k-1} \cdot \lambda}{(k-1)!} = \lambda \cdot \sum_{k=1}^{\infty} \frac{e^{-\lambda} \cdot \lambda^{k-1}}{(k-1)!} = \lambda.$$

$$\text{Var}(X) = E(X^2) - (EX)^2.$$

= \lambda.

$$\begin{aligned} \textcircled{2} E(X^2) &= \sum_{k=0}^{\infty} k^2 \cdot P(X=k) = \sum_{k=1}^{\infty} \frac{k^2 \cdot e^{-\lambda} \cdot \lambda^k}{k!} = e^{-\lambda} \cdot \lambda + \sum_{k=2}^{\infty} \frac{k \cdot e^{-\lambda} \cdot \lambda^k}{(k-1)!} \\ &= e^{-\lambda} \cdot \lambda + \sum_{k=2}^{\infty} \frac{e^{-\lambda} \cdot \lambda^k}{(k-2)!} + \sum_{k=2}^{\infty} \frac{e^{-\lambda} \cdot \lambda^{k-1}}{(k-1)!} \cdot \lambda \\ &= e^{-\lambda} \cdot \lambda + \lambda^2 + (1 - e^{-\lambda}) \cdot \lambda = \lambda^2 + \lambda. \end{aligned}$$

$$\textcircled{3} \text{Var}(X) = E(X^2) - (EX)^2 = \lambda.$$

Poisson Approximation

Cramer-Lundberg model.

Let $A_1, A_2, \dots, A_n$ be events with $p_j = P(A_j)$, where n is large, the p_j are small, and the A_j are independent or ~~weakly dependent~~. Let

$$X = \sum_{j=1}^n I(A_j)$$

count how many of the A_j occur. Then X is approximately Pois(λ), with $\lambda = \sum_{j=1}^n p_j$.

Example: Birthday Problem Revisited

1° m people; $\binom{m}{2}$ pairs, $j = 1, 2, \dots, \binom{m}{2}$.

Prob (each pair of people has the same birthday) = $\frac{1}{365}$.

2° A_j : "j-th pair of people has the same birthday". $P(A_j) = \frac{1}{365} = p$.
 $I_j = I(A_j)$; $n = \binom{m}{2}$; $X = \sum_j I_j$. # of birthday-match pairs.

3° Poisson approximation: $X \sim \text{Pois}(\lambda)$. $\lambda = np = \binom{m}{2} p$.

4° Prob. (at least one birthday match) = $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda}$

$$m = 23, \quad \lambda = \binom{23}{2} \cdot \frac{1}{365} = \frac{253}{365} \quad P(X \geq 1) = 1 - e^{-\lambda} \approx 0.5002$$

Poisson & Binomial

- Poisson $\implies$ Binomial: **conditioning**
- Binomial $\implies$ Poisson: **taking a limit**

Sum of Independent Poissons

$$\begin{aligned}
 P(X+Y=k) &\stackrel{\text{LOTP}}{=} \sum_{j=0}^k P(X+Y=k | X=j) P(X=j) \\
 &= \sum_{j=0}^k \underbrace{P(Y=k-j | X=j)} P(X=j) \\
 &= \sum_{j=0}^k P(Y=k-j) P(X=j) = \sum_{j=0}^k \frac{e^{-\lambda_2} \lambda_2^{k-j}}{(k-j)!} \cdot \frac{e^{-\lambda_1} \lambda_1^j}{j!}
 \end{aligned}$$

Theorem

If $X \sim \text{Pois}(\lambda_1)$, $Y \sim \text{Pois}(\lambda_2)$, and X is independent of Y , then $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$.

$$\begin{aligned}
 &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \cdot \sum_{j=0}^k \frac{k!}{(k-j)! \cdot j!} \cdot \lambda_1^j \cdot \lambda_2^{k-j} \\
 &= \frac{e^{-(\lambda_1 + \lambda_2)} \cdot (\lambda_1 + \lambda_2)^k}{k!} \quad \binom{k}{j} \cdot \lambda_1^j \lambda_2^{k-j}
 \end{aligned}$$

Poisson Given A Sum of Poissons

$$\begin{aligned}
 P(X=k | X+Y=n) &= \frac{P(X=k, X+Y=n)}{P(X+Y=n)} = \frac{P(X=k, Y=n-k)}{P(X+Y=n)} \\
 &= \frac{P(X=k)P(Y=n-k)}{P(X+Y=n)}.
 \end{aligned}$$

Theorem

If $X \sim \text{Pois}(\lambda_1)$, $Y \sim \text{Pois}(\lambda_2)$, and X is independent of Y , then the conditional distribution of X given $X + Y = n$ is $\text{Bin}(n, \lambda_1/(\lambda_1 + \lambda_2))$.

$$\begin{aligned}
 &= \frac{\frac{\cancel{e^{-\lambda_1}} \cdot \lambda_1^k}{k!} \cdot \frac{\cancel{e^{-\lambda_2}} \cdot \lambda_2^{n-k}}{(n-k)!}}{\frac{\cancel{e^{-(\lambda_1+\lambda_2)}} \cdot (\lambda_1+\lambda_2)^n}{n!}} \\
 &= \frac{n!}{(n-k)!k!} \cdot \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k} \\
 &\quad \left(\binom{n}{k} \cdot p^k (1-p)^{n-k}\right) \quad \text{where } p = \frac{\lambda_1}{\lambda_1+\lambda_2}
 \end{aligned}$$

Poisson Approximation to Binomial

1° $\lambda = np$. Given k , ($0 \leq k \leq n$). $X \sim \text{Bin}(n, p)$

$$P(X=k) = \binom{n}{k} \cdot p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}, \quad p = \lambda/n.$$

Theorem

If $X \sim \text{Bin}(n, p)$ and we let $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np$ remains fixed, then the PMF of X converges to the $\text{Pois}(\lambda)$ PMF. More generally, the same conclusion holds if $n \rightarrow \infty$ and $p \rightarrow 0$ in such a way that np converges to a constant λ . λ^k/n^k

$$= \frac{n(n-1) \cdots (n-k+1)}{k!} \cdot \left(\frac{\lambda}{n}\right)^k \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} = 1 \cdot \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \cdot \frac{\lambda^k}{k!} \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$= \frac{\lambda^k}{k!} \cdot \underbrace{\left[1 \cdot \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)\right]}_1 \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_1 \xrightarrow{n \rightarrow \infty} \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$$\xrightarrow{n \rightarrow \infty} \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda} \cdot 1 \sim \text{Pois}(\lambda)$$

Proof

Visitors to A Website

$$n=10^6, \quad p=2 \times 10^{-6}, \quad \lambda = np = 2.$$

$$P(Y=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$$Y: \text{ \# of visitors. } \quad Y \sim \text{Pois}(\lambda)$$

The owner of a certain website is studying the distribution of the number of visitors to the site. Every day, a million people independently decide whether to visit the site, with probability $p = 2 \times 10^{-6}$ of visiting. Give a good approximation for the probability of getting *at least three* visitors on a particular day.

$$\begin{aligned} P(Y \geq 3) &= 1 - P(Y=0) - P(Y=1) - P(Y=2) \\ &= 1 - 5e^{-2} \doteq 0.3233 \end{aligned}$$

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- 6 Distance between Two Probability Distributions**
- 7 Probability Generating Functions
- 8 Reading for Fun

Typical Distance Measures

- Total Variation Distance
- Kullback–Leibler Divergence
- Jensen–Shannon Divergence
- Bhattacharyya Distance
- Wasserstein Distance (or called “Kantorovich–Rubinstein”)

Total Variation Distance

- Distance measure between two probability distributions.
- Apply such measure to characterize the accuracy of Poisson approximation.

Definition

The total variation distance between two distributions μ and ν on a countable set Ω is

$$\begin{aligned} d_{TV}(\mu, \nu) &= \|\mu - \nu\|_{TV} \\ &= \frac{1}{2} \sum_{x \in \Omega} |\mu(x) - \nu(x)|. \end{aligned}$$

Example

$$\textcircled{1} \mu(1) = p, \mu(0) = 1-p, \mu(n) = 0, \forall n \geq 2.$$

$$\nu(n) = \frac{e^{-p} \cdot p^n}{n!}, n \geq 0.$$

$$= | -\mu(0) - \nu(1) | + \underbrace{\sum_{n=2}^{\infty} \nu(n)}$$

$$\textcircled{2} d_{TV}(\mu, \nu) = \sum_{n \in \Omega} |\mu(n) - \nu(n)| = |\mu(0) - \nu(0)| + |\mu(1) - \nu(1)| + \underbrace{\sum_{n=2}^{\infty} |\mu(n)|}$$

Let μ be the distribution with $\mu(1) = p$ and $\mu(0) = 1 - p$. Let ν be a Poisson distribution with mean p . Then we have $d_{TV}(\mu, \nu) \leq p^2$.

$$\begin{aligned} &= |1-p-e^{-p}| + |p-e^{-p}p| + (1-e^{-p}-e^{-p}p) \\ &= \underbrace{e^{-p}}_{p-1} + \underbrace{p(1-e^{-p})} + \underbrace{1-e^{-p}-e^{-p}p} \\ &= 2p(1-e^{-p}) \leq 2p^2 \end{aligned}$$

$$\textcircled{3} d_{TV}(\mu, \nu) \leq p^2$$

The Law of Small Numbers *The Law of Rare Events*

Theorem

Given independent random variables $Y_1, \dots, Y_n$ such that for any $1 \leq m \leq n$, $P(Y_m = 1) = \underline{p_m}$ and $P(Y_m = 0) = \underline{1 - p_m}$. Let $S_n = Y_1 + \dots + Y_n$. Suppose

$$\sum_{m=1}^n p_m \rightarrow \lambda \in (0, \infty) \quad \text{as } n \rightarrow \infty,$$

and

$$\max_{1 \leq m \leq n} p_m \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

then

$$d_{TV}(S_n, \text{Pois}(\lambda)) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Gap of Poisson Approximation

- A bound on the gap due to Hodges and Le Cam (1960):

$$d_{TV}(S_n, \text{Pois}(\lambda)) \leq \sum_{m=1}^n p_m^2,$$

- by Stein-Chen method (C.Stein 1987) we can have a tighter bound on the gap:

$$d_{TV}(S_n, \text{Pois}(\lambda)) \leq \min(1, \frac{1}{\lambda}) \sum_{m=1}^n p_m^2.$$

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- 8 Reading for Fun

Probability Generating Function

Definition

The *probability generating function* (PGF) of a nonnegative integer-valued r.v. X with PMF $p_k = P(X = k)$ is the generating function of the PMF. By LOTUS, this is

$$f(t) = E(t^X) = \sum_{k=0}^{\infty} p_k t^k. \quad = \sum_{k=0}^{\infty} P(X=k) \cdot t^k = \sum_{k=0}^{\infty} p_k t^k$$

The PGF converges to a value in $[-1, 1]$ for all t in $[-1, 1]$ since $\sum_{k=0}^{\infty} p_k = 1$ and $|p_k t^k| \leq p_k$ for $|t| \leq 1$.

Example: Generating Dice Probabilities

① $E[t^X] = \sum_{k=0}^{\infty} P(X=k)t^k = \sum_{k=0}^{\infty} p_k t^k$ is polynomial function of t ,

② $E[XY] = E[X]E[Y]$
if X, Y independent.

the coefficient of t^{18} : $P(X=18)$

t^{19} : $P(X=19)$

t^{20} : $P(X=20)$

$$E[XY] = \sum_x \sum_y xy P(X=x) P(Y=y) = \sum_x x P(X=x) \cdot \sum_y y P(Y=y) = E[X]E[Y]$$

Let X be the total from rolling 6 fair dice, and let $X_1, \dots, X_6$ be the individual rolls. What is $P(X=18)$? $X = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$.

③ $E[t^X] = E[t^{X_1}] \cdot E[t^{X_2}] \cdots E[t^{X_6}]$

$$6 \leq X \leq 36$$

$$= (E[t^{X_1}])^6$$

$$E[t^{X_1}] = \sum_{j=1}^6 P(X_1=j) \cdot t^j = \frac{1}{6}(t + t^2 + \cdots + t^6)$$

$$P(X=18)$$

$$= \frac{3431}{6^6}$$

④ $E[t^X] = (E[t^{X_1}])^6 = \frac{t^6}{6^6} (1 + t + \cdots + t^5)^6$

Solution

PGF and Moments

$$\textcircled{1} g(t) = \sum_{k=0}^{\infty} p_k t^k = p_0 + \sum_{k=1}^{\infty} p_k t^k ; \quad \underline{g(t) = p_1 + \sum_{k=2}^{\infty} p_k t^{k-1} : k}$$

$$g'(t)|_{t=1} = \sum_{k=1}^{\infty} p_k \cdot k + 0 \cdot p_0 = E(X).$$

Let X be a nonnegative integer-valued r.v. with PMF $p_k = P(X = k)$, and the PGF of X is $\underline{g(t) = \sum_{k=0}^{\infty} p_k t^k}$, we have

- $\underline{E(X) = g'(t)|_{t=1}}$, ✓

- $\underline{E(X(X-1)) = g''(t)|_{t=1}}$. $E[X^2] - E[X]$. $\text{Var}(X) = E[X^2] - (EX)^2$

$$\textcircled{2} g''(t) = \sum_{k=2}^{\infty} k(k-1) \cdot p_k \cdot t^{k-2}, \quad g''(t)|_{t=1} = \sum_{k=2}^{\infty} k(k-1) \cdot p_k + 0 \cdot p_1 + 0 \cdot p_0$$

$$\textcircled{3} g''(t)|_{t=1} = E[X(X-1)].$$

PGF and Moments

$$1^{\circ} g(t) = \sum_{k=0}^{\infty} p_k t^k = p_0 + \sum_{k=1}^{\infty} p_k t^k; \quad P(X=0) = g(0) = p_0,$$

$$2^{\circ} g'(t) = p_1 + \sum_{k=2}^{\infty} p_k t^{k-1} \cdot k; \quad P(X=1) = g'(0) = p_1,$$

$$3^{\circ} g''(t) = 2p_2 + \sum_{k=3}^{\infty} k \cdot (k-1) \cdot p_k t^{k-2}; \quad P(X=2) = \frac{g''(0)}{2}.$$

$$4^{\circ} g^{(m)}(t) = m! \cdot p_m + \sum_{k=m+1}^{\infty} \frac{k!}{(k-m)!} \cdot t^{k-m} \cdot p_k.$$

$$P(X=m) = \frac{g^{(m)}(0)}{m!}.$$

PGF and Moments

Binomial PMF

i.i.d.

$$① X \sim \text{Bin}(n, p), \quad X = X_1 + \dots + X_n, \quad X_i \sim \text{Bern}(p), \quad (1 \leq i \leq n).$$

$$② g_X(t) = E[t^X] = E[t^{X_1 + X_2 + \dots + X_n}] = E[t^{X_1}] \dots E[t^{X_n}] = (E[t^{X_1}])^n$$

$$③ E[t^{X_i}] = t^0 \cdot (1-p) + t^1 \cdot p = pt + q, \quad q = (1-p).$$

$$④ g_X(t) = (pt + q)^n = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} t^k. \quad \left| \quad p(X=k) = \binom{n}{k} p^k q^{n-k} \right.$$

$$⑤ p(X=0) = g_X(0) = q^n, \quad p(X=1) = g_X'(1) = npq^{n-1}$$

$$p(X=k) = \frac{g_X^{(k)}(0)}{k!} = \binom{n}{k} \cdot p^k q^{n-k}$$

Binomial Moments

$$\underline{g_X(t)} = (pt + q)^n, \quad p + q = 1.$$

$$g'_X(t) = np(pt + q)^{n-1}, \quad g'_X(t)|_{t=1} = np = E[X].$$

$$g''_X(t) = n(n-1) \cdot p^2 (pt + q)^{n-2}, \quad g''_X(t)|_{t=1} = n(n-1)p^2 = \underline{E[X(X-1)]}$$

$$\text{Var}(X) = g''_X(t)|_{t=1} + g'_X(t)|_{t=1} - (g'_X(t)|_{t=1})^2$$

$$\underline{E\left[\binom{X}{k}\right] = \binom{n}{k} p^k, \quad k \geq 2}$$

$$H = p, T: 1-p$$

Example: Pattern Matching

$$\textcircled{1} p_k = P(N=k); p_0=0; p_1=0; p_2=p^2; p_3=(1-p)p^2=qp^2$$

$$N=4 \begin{matrix} & 1 & 2 & 3 & 4 \\ & H & T & H & H \\ < & T & T & H & H \end{matrix}$$

$\textcircled{2}$ First-step method.

Suppose a coin with probability p for heads is tossed repeatedly, and we obtain a sequence of H and T (H denotes Head and T denotes Tail). Let N denote the number of toss to observe the first occurrence of the pattern "HH". Find $E(N)$ and $\text{Var}(N)$.

$$k \geq 3, S_1: \text{result of first toss}, S_1 = H \text{ or } T.$$

$$\begin{aligned} \text{LOTP: } P_k = P(N=k) &= P(N=k | S_1=H) P(S_1=H) + P(N=k | S_1=T) P(S_1=T) \\ &= P(N=k, S_1=H) + P(N=k, S_1=T). \end{aligned}$$

Example: Pattern Matching

 S_2 = result of the 2nd toss

$$\textcircled{3} P(N=k, S_1=H) = P(S_1=H) P(S_2=T) \cdot P(N=k-2) = p \cdot q P_{k-2}$$

$$P(N=k, S_1=T) = P(S_1=T) P(N=k-1) = q P_{k-1}.$$

$$\textcircled{4} P_k = P(N=k, S_1=H) + P(N=k, S_1=T)$$

$$P_k = p q P_{k-2} + q P_{k-1}, \quad k \geq 3$$

$$P_0 = 0, \quad P_1 = 0, \quad P_2 = p^2,$$

method 1:

sequence: find P_k .

$$E[N] = \sum_{k=0}^{\infty} k \cdot P_k.$$

method 2: PGF $\rightarrow E[N]$

Example: Pattern Matching

$$\begin{cases} p_k = p \mathbb{I} p_{k-2} + \mathbb{I} p_{k-1}, & k \geq 3. \\ p_0 = 0, & p_1 = 0, & p_2 = p^2. \end{cases}$$

$$\textcircled{5} \text{ PGF of } N. \quad g(t) = E[t^N] = \sum_{k=0}^{\infty} p_k t^k = \sum_{k=1}^{\infty} p_k t^k = \sum_{k=2}^{\infty} p_k t^k$$

$$= p_2 t^2 + \sum_{k=3}^{\infty} p_k t^k = p^2 t^2 + \sum_{k=3}^{\infty} p_k t^k = g(t).$$

on the other hand, $p_k = p \mathbb{I} p_{k-2} + \mathbb{I} p_{k-1}$, $k \geq 3$.

$$\begin{aligned} \sum_{k=3}^{\infty} p_k t^k &= \sum_{k=3}^{\infty} (\mathbb{I} p_{k-1} + p \mathbb{I} p_{k-2}) t^k = \sum_{k=3}^{\infty} \mathbb{I} p_{k-1} t^k + \sum_{k=3}^{\infty} p \mathbb{I} p_{k-2} t^k \\ &= \mathbb{I} t \sum_{k=3}^{\infty} p_{k-1} t^{k-1} + p \mathbb{I} t^2 \sum_{k=3}^{\infty} p_{k-2} t^{k-2} \\ &= \mathbb{I} t \sum_{k=2}^{\infty} p_k t^k + p \mathbb{I} t^2 \sum_{k=1}^{\infty} p_k t^k \\ &= \mathbb{I} t g(t) + p \mathbb{I} t^2 g(t) = (\mathbb{I} t + p \mathbb{I} t^2) g(t) \end{aligned}$$

Example: Pattern Matching

$$\Rightarrow g(t) = \frac{p^2 t^2}{1 - st - p s t^2} \quad \text{PGF of } N.$$

$$E[N] = g'(t) \big|_{t=1} = g'(1) = \frac{1}{p} + \frac{1}{p^2}$$

$$\text{Var}[N] = g''(1) + g'(1) - [g'(1)]^2 = \frac{1 - p^5 - 5 p s^4}{s^2 p^4}$$

“H H H”

Example: Pattern Matching

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Probability Method

- Paul Erdős initiated this method: Erdős Method.
- Widely used in information theory & combinatorics & theoretical computer science.
- Main idea: to prove the existence of a structure with certain properties using probability or expectation.

Principle I

- First we construct an appropriate probability space of structures.
- Then we show that a randomly chosen element in this space has the desired properties with positive probability.

Theorem (The Possibility Principle)

Let A be the event that a randomly chosen object in a collection has a certain property. If $P(A) > 0$, then there exists an object with such property.

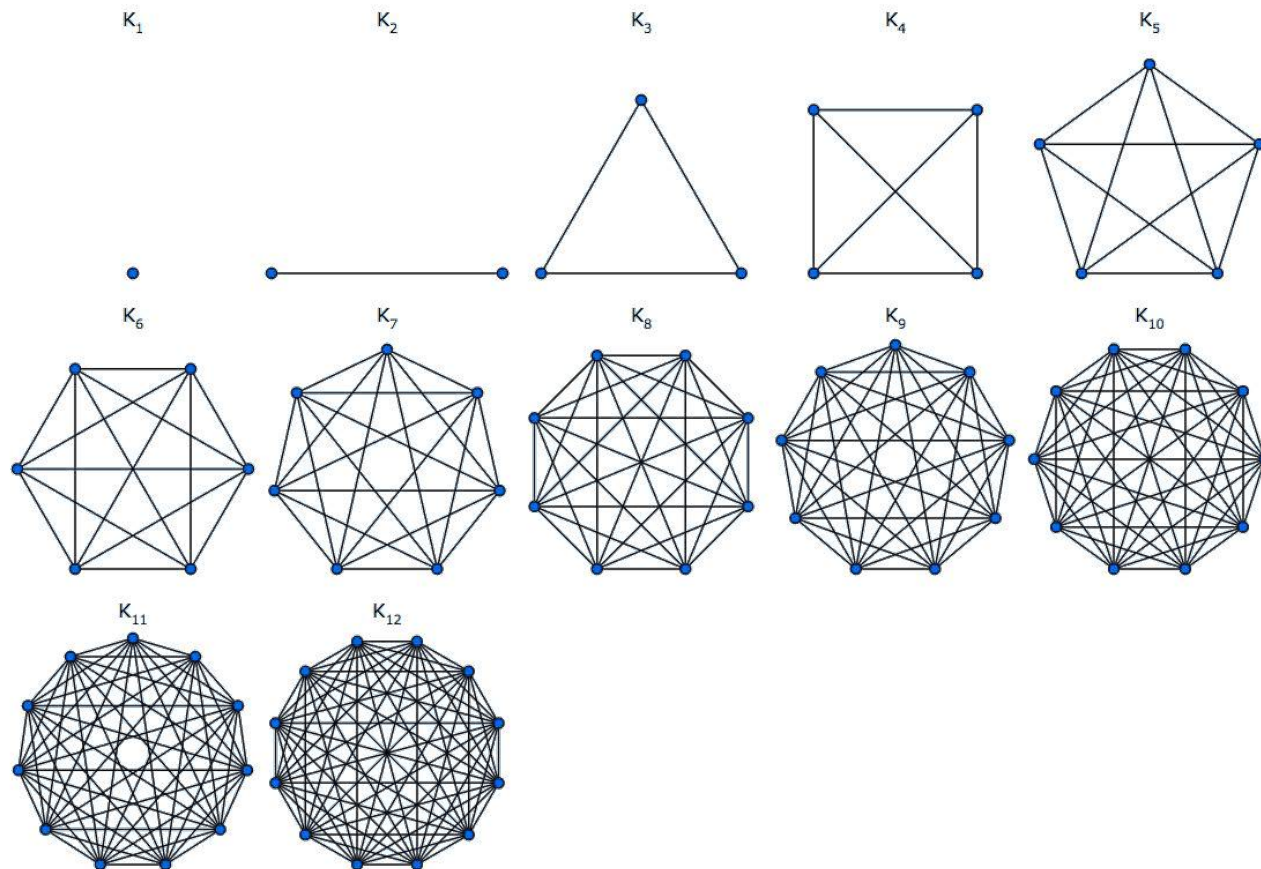
Principle II

Theorem (The Good Score Principle)

Let X be the score of a randomly chosen object. If $E(X) \geq c$, then there exists an object with a score of at least c .

Example: Graph Coloring

- Complete graph (clique): a simple undirected graph in which every pair of distinct vertices is connected by a unique edge.
- Complete graph K_n : a graph with n nodes and $\binom{n}{2}$ edges.



Example: Graph Coloring

Theorem

Given a complete graph K_n ($n \geq 3$), if $\binom{n}{m} 2^{-\binom{m}{2}+1} < 1$, then it is possible to color the edges of K_n with two colors so that it has no monochromatic K_m subgraph ($1 < m < n$).

Testing Polynomial Identities

- Randomized algorithms can be dramatically more efficient than their best known deterministic counterparts.
- Input two polynomials Q and R over n variables, with coefficients in some field, and decides whether $Q \equiv R$.
- Example: $Q(x_1, x_2) = (1 + x_1)(1 + x_2)$, $R(x_1, x_2) = 1 + x_1 + x_2 + x_1x_2$.
- n -variable polynomial $\prod_{i=1}^n (x_i + x_{i+1})$ expands into $O(2^n)$ monomials.

The Schwartz-Zippel Algorithm

- A Monte Carlo algorithm with a bounded probability of false positive and no false negative.
- Input polynomial $M(x_1, \dots, x_n)$ and test whether $M \equiv 0 (M = Q - R)$.
- Assign values $r_1, \dots, r_n$ chosen independently and uniformly at random from a finite set S to $x_1, \dots, x_n$.
- Test if $M(r_1, \dots, r_n) = 0$, outputting “Yes” if so and “No” otherwise.
- If “No”, then $M \not\equiv 0$.
- if “Yes”, it is possible that $M \not\equiv 0$ but $r_1, \dots, r_n$ happens to be a zero of M .

Schwartz-Zippel Lemma

Lemma

Let $M \in F(x_1, x_2, \dots, x_n)$ be a non-zero polynomial of total degree $d \geq 0$ over a field F . Let S be a finite subset of F and let $r_1, r_2, \dots, r_n$ be selected at random independently and uniformly from S . Then

$$P[M(r_1, r_2, \dots, r_n) = 0] \leq \frac{d}{|S|}.$$

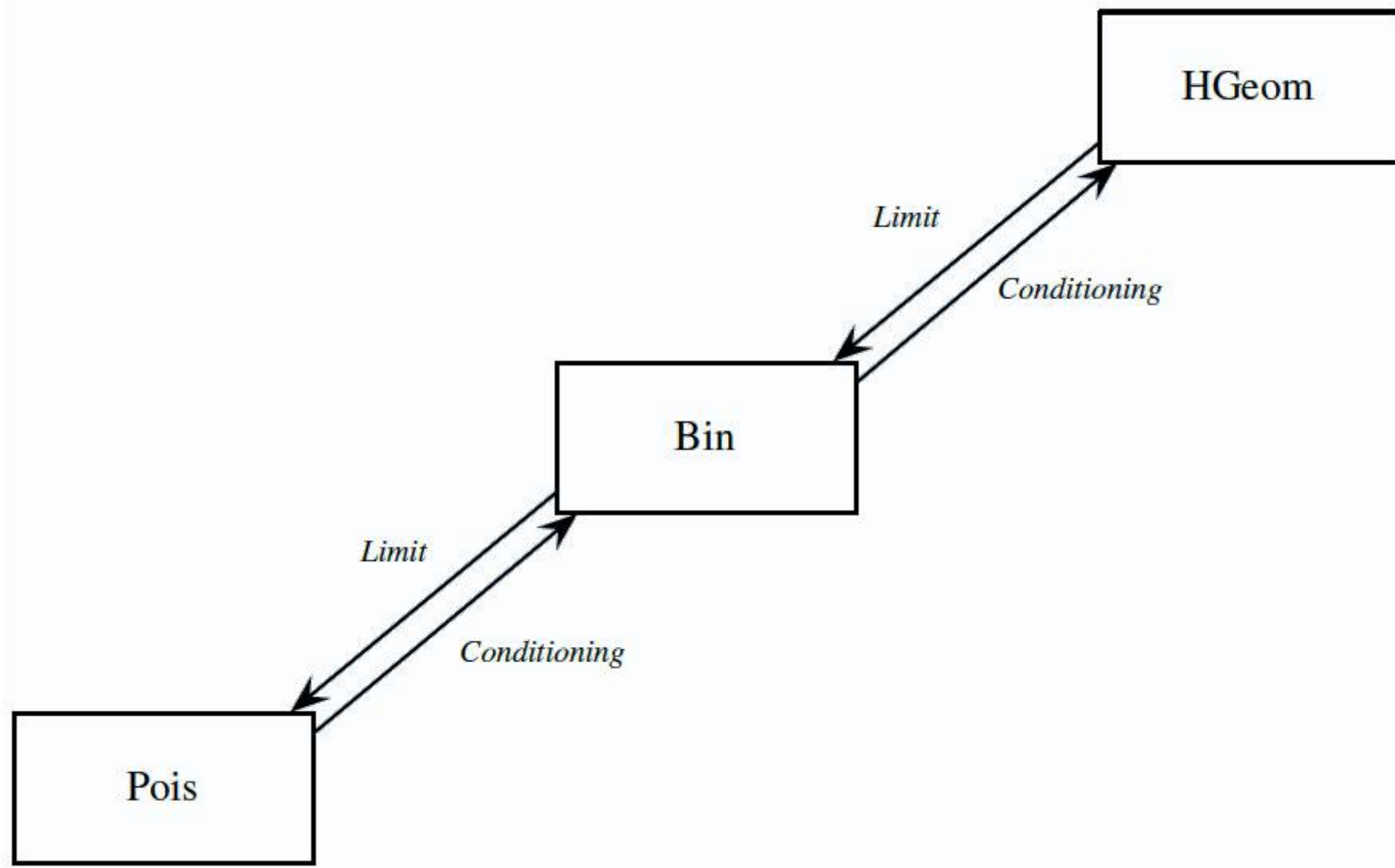
Remarks

- If we take the set S to have cardinality at least twice the degree of our polynomial ($|S| \geq 2d$), we can bound the probability of error (false positive) by $1/2$.
- This can be reduced to any desired small number by repeated trials.

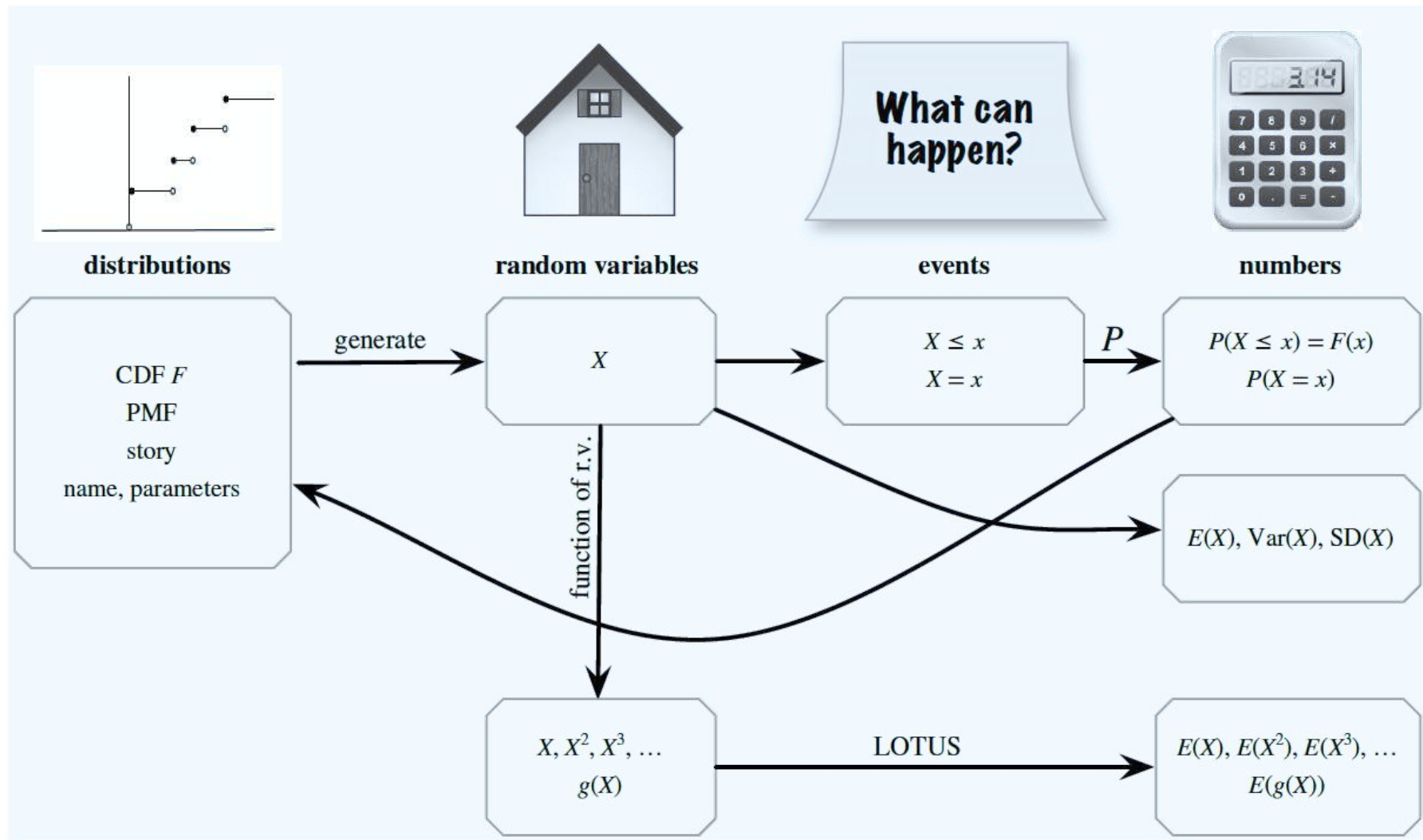
Summary 1

	With replacement	Without replacement
Fixed number of trials	Binomial	Hypergeometric
Fixed number of successes	Negative Binomial	Negative Hypergeometric

Summary 2



Summary 3



References

- Chapters 4 & 6 of **BH**
- Chapter 2 of **BT**